

mb-pc40s-2014-jan

Typst conversion review. OCR content and mathematical accuracy require editorial review.

Question 1

Find the coterminal angle to $\frac{27\pi}{5}$ over the interval $[-360^\circ, 0^\circ)$.

Solution

Method 1

$$\begin{aligned}\frac{27\pi}{5} \left(\frac{180^\circ}{\pi} \right) &= 27(36^\circ) \\ &= 972^\circ \\ 972^\circ - (360^\circ)(3) &= -108^\circ\end{aligned}$$

1 mark for conversion to degrees

$\frac{1}{2}$ mark for coterminal angle

$\frac{1}{2}$ mark for correct domain

2 marks

Method 2

$$\begin{aligned}-\frac{3\pi}{5} \text{ is a coterminal angle to } \frac{27\pi}{5} \\ -\frac{3\pi}{5} \cdot \frac{180^\circ}{\pi} = -108^\circ\end{aligned}$$

$\frac{1}{2}$ mark for coterminal angle

$\frac{1}{2}$ mark for correct domain

1 mark for conversion to degrees

2 marks

Question 2

Solve the following equation over the interval $0 \leq \theta < 2\pi$.

$$(\tan \theta - 3)(\tan \theta + 1) = 0$$

Solution

$$\tan \theta = 3$$

$$\tan \theta = -1$$

$$\theta_r = 1.249046$$

$$\theta_r = \frac{\pi}{4}$$

$$\theta = 1.249046$$

$$\theta = 4.390639$$

$$\theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

or

$$\theta = 1.249, 4.391$$

$$\theta = 2.356194$$

$$\theta = 5.497787$$

$$\therefore \theta = 1.249, \frac{3\pi}{4}, 4.391, \frac{7\pi}{4}$$

or

$$\theta = 1.249, 2.356, 4.391, 5.498$$

1 mark ($\frac{1}{2}$ mark for each branch)

2 marks ($\frac{1}{2}$ mark for each value of θ)

3 marks

Question 3

An earthquake in Vancouver had a magnitude of 6.3 on the Richter scale. An earthquake in Japan had a magnitude of 8.9 on the Richter scale.

How many times more intense was the Japan earthquake than the Vancouver earthquake?

You may use the formula below:

$$M = \log\left(\frac{A}{A_0}\right)$$

where M is the magnitude of the earthquake on the Richter scale

A is the intensity of the earthquake

A_0 is the intensity of a standard earthquake

Express your answer as a whole number.

Solution

Method 1

Vancouver: substitute $M = 6.3$

$$6.3 = \log\left(\frac{A}{A_0}\right)$$

$$10^{6.3} = \frac{A}{A_0}$$

$$A = 10^{6.3} A_0$$

$\frac{1}{2}$ mark for exponential form

Japan: substitute $M = 8.9$

$$8.9 = \log\left(\frac{A}{A_0}\right)$$

$$10^{8.9} = \frac{A}{A_0}$$

$$A = 10^{8.9} A_0$$

$\frac{1}{2}$ mark for exponential form

To compare the two earthquakes divide their intensities.

$$\begin{aligned}\frac{\text{the intensity of Japan}}{\text{the intensity of Vancouver}} &= \frac{10^{8.9} A_0}{10^{6.3} A_0} \\ &= 398.107 \\ &= 398\end{aligned}$$

1 mark for comparison

2 marks

Question 4

Find and simplify the last term in the expansion of $(2y - 3x)^7$.

Solution

$$\begin{aligned}t_8 &= {}_7C_7(2y)^0(-3x)^7 \\ &= -2187x^7\end{aligned}$$

1/2 mark for ${}_7C_7$
1/2 mark for $(2y)^0$

1 mark for $(-3x)^7$ which is equal to $-2187x^7$
2 marks

Note(s):

- no deduction if ${}_7C_7$ and $(2y)^0$ are not shown

Question 5

Given $\log_a 9 = 1.129$ and $\log_a 4 = 0.712$, find the value of $\log_a 12$.

Solution

Method 1

$$\log_a 9 = 1.129$$

$$\log_a 3^2 = 1.129$$

$$2 \log_a 3 = 1.129$$

$$\log_a 3 = 0.5645$$

1 mark for power rule

$$\log_a 12 = \log_a (4 \cdot 3)$$

$$= \log_a 4 + \log_a 3$$

$$= 0.712 + 0.5645$$

$$= 1.2765$$

$$= 1.277$$

1 mark for writing 12 as a product

1 mark for product rule

3 marks

Method 2

$$\log_a 12 = \log_a (\sqrt{9} \cdot 4)$$

$$= \frac{1}{2} \log_a 9 + \log_a 4$$

$$= \frac{1}{2}(1.129) + 0.712$$

$$= 1.2765$$

$$= 1.277$$

1 mark for writing 12 as a product

1 mark for power rule

1 mark for product rule

3 marks

Method 3

$$\log_a 9 = 1.129$$

$$a^{1.129} = 9$$

$$a = 9^{\frac{1}{1.129}}$$

$$a = 7$$

1 mark for exponential form

1 mark for solving for a

1 mark for value of $\log_a 12$

3 marks

Question 6

(initi)

How many different ways can 4 girls and 4 boys be arranged in a row if the girls and the boys must alternate?

Solution

Method 1

$$\begin{array}{cccccccccccc} & 8 & \cdot & 4 & \cdot & 3 & \cdot & 3 & \cdot & 2 & \cdot & 2 & \cdot & 1 & \cdot & 1 \\ \text{choose} & \uparrow & & \uparrow & & \underbrace{\hspace{2cm}} & & & & & & & & & & \\ \text{one} & & & & & \text{choose} & & & & & \text{alternate the rest} & & & & & \\ \text{person} & & & & & \text{the} & & & & & & & & & & \\ & & & & & \text{other} & & & & & & & & & & \\ & & & & & \text{gender} & & & & & & & & & & \end{array} = 1152 \text{ ways}$$

1 mark for beginning with either gender

1 mark for alternating genders

2 marks

Method 2

Case 1: $4 \cdot 4 \cdot 3 \cdot 3 \cdot 2 \cdot 2 \cdot 1 \cdot 1 = 576$

$\bar{B}\bar{G}\bar{B}\bar{G}\bar{B}\bar{G}\bar{B}\bar{G}$

1 mark for arrangement of alternating genders

Case 2:

$$\frac{4}{G} \cdot \frac{4}{B} \cdot \frac{3}{G} \cdot \frac{3}{B} \cdot \frac{2}{G} \cdot \frac{2}{B} \cdot \frac{1}{G} \cdot \frac{1}{B} = 576$$

$\frac{1}{2}$ mark for two cases

$\frac{1}{2}$ mark for addition of cases

2 marks

Question 7

Solve the following equation over the interval $[0, 2\pi]$.

$$2 \cos 2\theta - 1 = 0$$

Solution

Method 1

$$\begin{aligned}2 \cos 2\theta - 1 &= 0 \\2(2 \cos^2 \theta - 1) - 1 &= 0 \\4 \cos^2 \theta - 3 &= 0 \\\cos^2 \theta &= \frac{3}{4} \\\cos \theta &= \pm \frac{\sqrt{3}}{2} \\\theta_r &= \frac{\pi}{6} \\\theta &= \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}\end{aligned}$$

1 mark for identity

1 mark for solving for $\cos \theta$

2 marks for solutions ($\frac{1}{2}$ mark for each consistent solution)

4 marks

Method 2

نص

Find all zeros over the interval $[0, 2\pi]$.

$$\theta = 0.524, 2.618, 3.665, 5.760$$

1 mark for equation

1 mark for the graph with zeroes

indicated

2 marks for solutions

4 marks

Method 3

$$\begin{aligned}2 \cos 2\theta - 1 &= 0 \\\cos 2\theta &= \frac{1}{2} \\\text{let } x &= 2\theta \\\therefore \cos x &= \frac{1}{2} \\x_r &= \frac{\pi}{3} \\x &= \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3} \\\therefore 2\theta &= \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3} \\\theta &= \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}\end{aligned}$$

$\frac{1}{2}$ mark for solving for $\cos 2\theta$

$\frac{1}{2}$ mark for reference angle

1 mark for correct angles

1 mark for coterminal angles

1 mark for all solutions over the interval $[0, 2\pi]$.

4 marks

Note(s):

- in Method 1, deduct a maximum of 1 mark if missing $\cos \theta = -\frac{\sqrt{3}}{2}$

Question 8

Alex incorrectly explains to Rashid that the graph of $y = 2f(x) + 5$ means you first move the graph of $y = f(x)$ up 5 units and then multiply the y values by 2 .

Explain to Rashid the correct way to transform the graph.

Solution

Alex explains the transformations correctly, but not in the correct order.

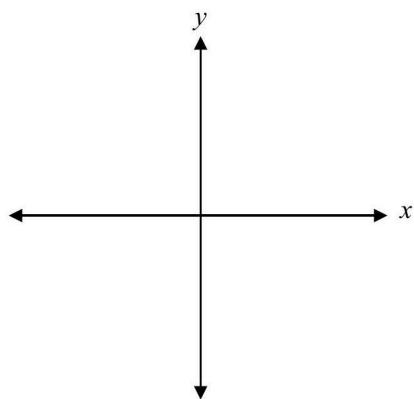
First multiply the y -values by 2 , then move the graph up 5 units.

1 mark for explanation

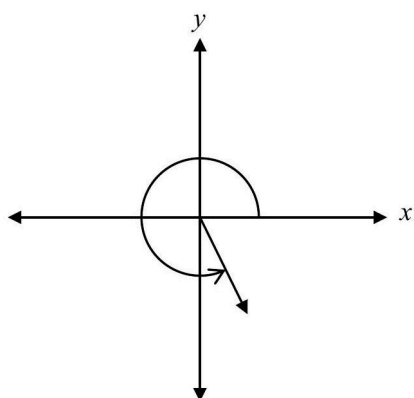
1 mark

Question 9

Sketch the angle of 5 radians in standard position.



Solution

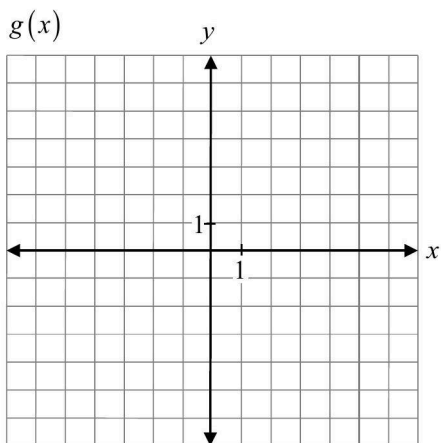
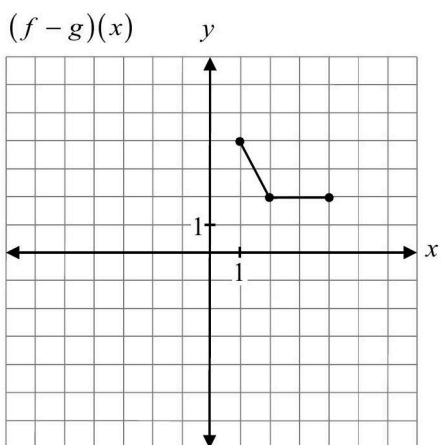
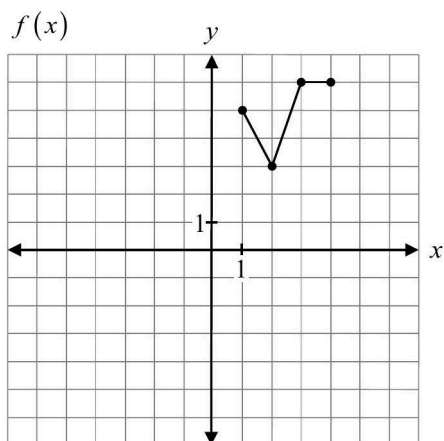


1 mark for angle drawn in Quadrant IV

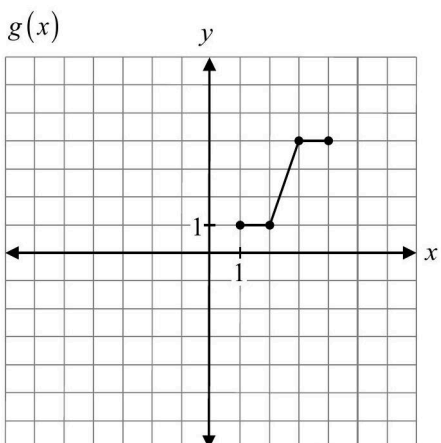
1 mark

Question 10

Given the graphs of $f(x)$ and $(f - g)(x)$, sketch the graph of $g(x)$.



Solution



1 mark for subtraction of $f(x) - (f - g)(x)$

1 mark for shape representing the operation

given

2 marks

Question 11

A particular math class has a large number of students. From this class, you are to create a committee of 4 students that has at least 1 girl.

Without actually solving the problem, explain the strategy you would use to find the total number of ways to select this committee.

Solution

Method 1

Find all ways to form the committee and subtract the number of ways with no girls on the committee.

1 mark for identifying cases

1 mark for operation on these cases

2 marks

Method 2

Find all scenarios with girls on the committee: 1 girl on the committee, 2 girls on the committee, 3 girls on the committee, and a committee with all girls. Then add all scenarios together to get the total number of ways to create a committee with at least 1 girl.

1 mark for identifying cases

1 mark for operation on these cases

2 marks

Note(s):

- in Method 2, deduct $\frac{1}{2}$ mark if one of the cases is missing

Question 12

a) Prove the identity below for all permissible values of θ .

$$\frac{1 + 2 \cos^2 \theta}{\cos^2 \theta} = \tan^2 \theta + 3$$

Left-Hand Side	Right-Hand Side

b) Determine all the non-permissible values for θ .

Solution

Method 1

a)

LHS	RHS
$\frac{1}{\cos^2 \theta} + \frac{2 \cos^2 \theta}{\cos^2 \theta}$ $\sec^2 \theta + 2$ $\tan^2 \theta + 1 + 2$ $\tan^2 \theta + 3$	$\tan^2 \theta + 3$
$\therefore \text{LHS} = \text{RHS}$	

1 mark for appropriate algebraic strategy

1 mark for appropriate identity substitutions

2 marks

b)

$$\begin{aligned}\cos^2 \theta &= 0 \\ \cos \theta &= 0 \\ \theta &= \frac{\pi}{2}, \frac{3\pi}{2}, \dots\end{aligned}$$

$\frac{1}{2}$ mark for $\cos^2 \theta = 0$

$\frac{1}{2}$ mark for any non-permissible value of θ

$\therefore$ the non-permissible values of θ

$$\begin{aligned}\text{are } \frac{\pi}{2} + k\pi, k \in \mathbb{I} \\ \text{or} \\ 90^\circ + 180k, k \in \mathbb{I}.\end{aligned}$$

1 mark for all non-permissible values of θ

2 marks

Method 2

a)

LHS	RHS
-----	-----

$\frac{1+2\cos^2\theta}{\cos^2\theta}$	$\tan^2\theta + 3$ $\sec^2\theta - 1 + 3$ $\sec^2\theta + 2$ $\frac{1}{\cos^2\theta} + 2$ $\frac{1}{\cos^2\theta} + \frac{2\cos^2\theta}{\cos^2\theta}$ $\frac{1+2\cos^2\theta}{\cos^2\theta}$

$$\therefore \text{LHS} = \text{RHS}$$

1 mark for appropriate identity substitutions

1 mark for appropriate algebraic strategy

2 marks

b)

$$\cos^2\theta = 0$$

$$\cos\theta = 0$$

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$\therefore$ the non-permissible values of θ

$$\text{are } \frac{\pi}{2} + k\pi, k \in \mathbb{I}$$

or

$$90^\circ + 180k, k \in \mathbb{I}$$

$\frac{1}{2}$ mark for $\cos^2\theta = 0$

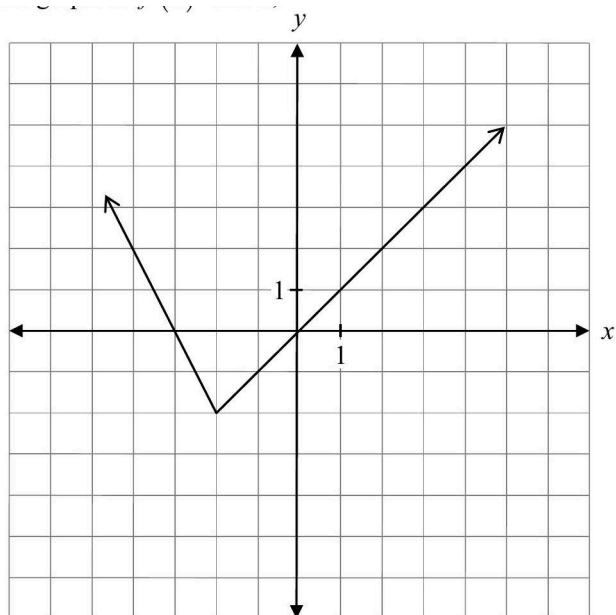
$\frac{1}{2}$ mark for any non-permissible value of θ

1 mark for all non-permissible values of θ

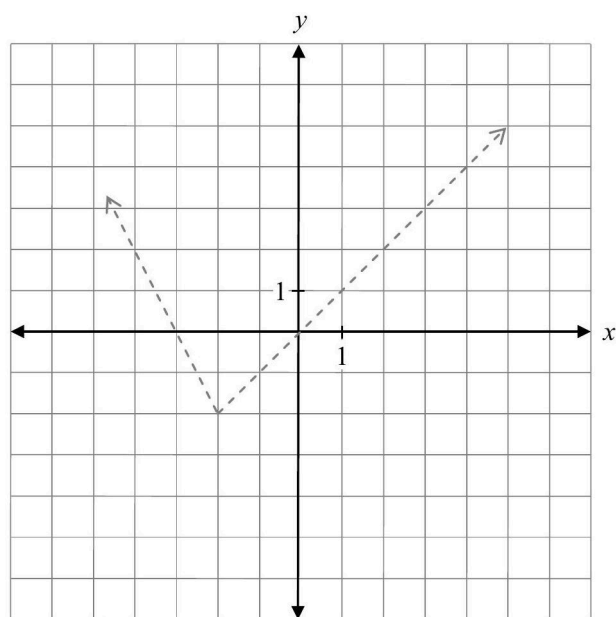
2 marks

Question 13

Given the graph of $f(x)$ below,



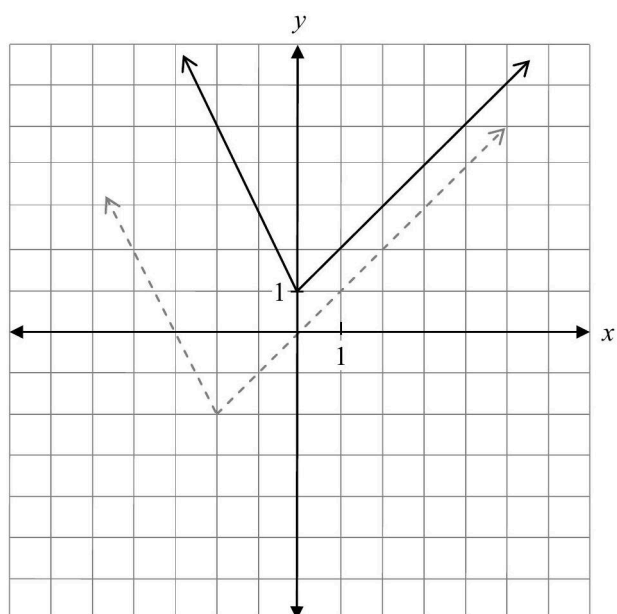
sketch the graph of $g(x) = f(x - 2) + 3$.



The graph of
 $f(x)$ has
already been
drawn for your
reference.

No marks will
be awarded for
the graph of
 $f(x)$.

Solution



1 mark for horizontal shift

1 mark for vertical shift

2 marks

Question 14

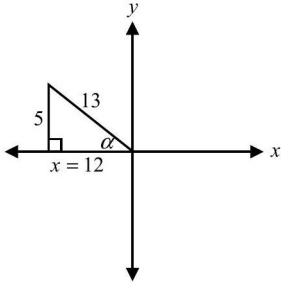
Given that $\sin \alpha = \frac{5}{13}$, where α is in Quadrant II, and $\cos \beta = \frac{2}{5}$, where β is in Quadrant IV,

find the exact value of:

a) $\cos(\alpha + \beta)$

b) $\sin 2\alpha$

Solution



$$x^2 + y^2 = r^2$$

$$x^2 + 25 = 169$$

$$x^2 = 144$$

$$x = \pm 12$$

$$x = -12$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$= \left(-\frac{12}{13}\right)\left(\frac{2}{5}\right) - \left(\frac{5}{13}\right)\left(-\frac{\sqrt{21}}{5}\right)$$

$$= -\frac{24}{65} + \frac{5\sqrt{21}}{65}$$

$$= \frac{5\sqrt{21} - 24}{65}$$

b)

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$= 2\left(\frac{5}{13}\right)\left(-\frac{12}{13}\right)$$

$$= -\frac{120}{169}$$

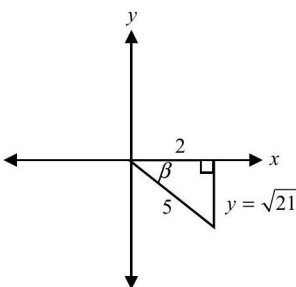
$$x^2 + y^2 = r^2$$

$$4 + y^2 = 25$$

$$y^2 = 21$$

$$y = \pm\sqrt{21}$$

$$y = -\sqrt{21}$$



$\frac{1}{2}$ mark for value of x

$\frac{1}{2}$ mark for value of y

$\frac{1}{2}$ mark for $\cos \alpha$

$\frac{1}{2}$ mark for $\sin \beta$

1 mark for substitution into correct identity

3 marks

1 mark for substitution into correct identity

1 mark

Note(s):

- accept any of the following values for x : $x = \pm 12, x = -12$ or $x = 12$
- accept any of the following values for y : $y = \pm\sqrt{21}, y = -\sqrt{21}$ or $y = \sqrt{21}$

Question 15

If $f(x) = x^3$ and $g(x) = 2x - 3$, what is the value of $\left(\frac{f}{g}\right)(-1)$?

Solution

$$\begin{aligned}f(-1) &= (-1)^3 \\&= -1\end{aligned}$$

$\frac{1}{2}$ mark for substituting into $f(x)$ and $g(x)$

$$\begin{aligned}g(-1) &= 2(-1) - 3 \\&= -5\end{aligned}$$

$$\begin{aligned}\left(\frac{f}{g}\right)(-1) &= \frac{-1}{-5} \\&= \frac{1}{5}\end{aligned}$$

$\frac{1}{2}$ mark for evaluating $\left(\frac{f}{g}\right)(-1)$

1 mark

Question 16

If the point $(4, -3)$ lies on the graph of $f(x)$, which point must lie on the graph of $2f(2x)$?

A.

$(8, -6)$

B.

$(2, -6)$

C.

$(8, -\frac{3}{2})$

D.

$(2, -\frac{3}{2})$

Solution

Correct answer: B

$(2, -6)$

Question 17

The graph of $y = \log_2(2x + 6)$ intersects the graph of $y = 4$ at:

A.

$$x = -1$$

B.

$$x = 1$$

C.

$$x = 5$$

D.

$$x = 14$$

Solution

Correct answer: C

$$x = 5$$

Question 18

Given the point $A(-3, 5)$ on the terminal arm of an angle θ , identify the value of $\cot \theta$.

A.

$$-\frac{3}{5}$$

B.

$$-\frac{5}{3}$$

C.

$$-\frac{4}{5}$$

D.

$$-\frac{5}{4}$$

Solution

Correct answer: A

$$-\frac{3}{5}$$

Question 19

The graph of $y = \left(\frac{1}{2}\right)^x$ compared to the graph of $x = \left(\frac{1}{2}\right)^y$ is a:

A.

reflection in the x -axis

B.

reflection in the y -axis

C.

reflection in the line $y = x$

D.

reciprocal function

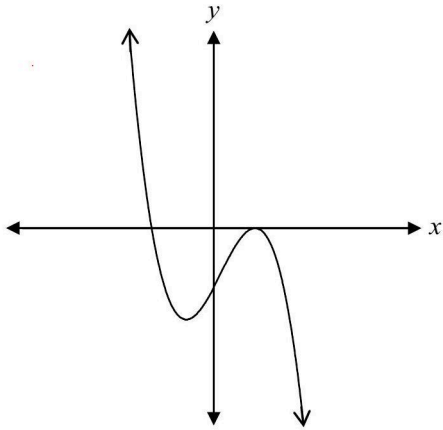
Solution

Correct answer: C

reflection in the line $y = x$

Question 20

1 mark



Given the above graph of a polynomial function, which one of the following statements can be true?

A.

The function has a degree of 4 with a positive leading coefficient.

B.

The function has a degree of 4 with a negative leading coefficient.

C.

The function has a degree of 3 with a positive leading coefficient.

D.

The function has a degree of 3 with a negative leading coefficient.

Solution

Correct answer: D

The function has a degree of 3 with a negative leading coefficient.

Question 21

Given that $(x + 3)$ is a factor of polynomial $P(x)$, which of the following is true?

A.

$$P(-3) = 0$$

B.

$$P(0) = -3$$

C.

$$P(0) = 3$$

D.

$$P(3) = 0$$

Solution

Correct answer: A

$$P(-3) = 0$$

Question 22

Which of the following is a reasonable estimate for the value of $\log 350$?

A.

2

B.

2.5

C.

2.8

D.

3

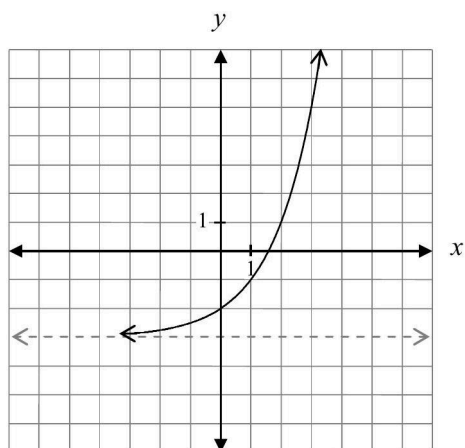
Solution

Correct answer: B

2.5

Question 23

The graph of the function $f(x)$ shown below is best described by the equation:



A.

$$f(x) = 2^{x+3}$$

B.

$$f(x) = 2^x + 3$$

C.

$$f(x) = 2^{x-3}$$

D.

$$f(x) = 2^x - 3$$

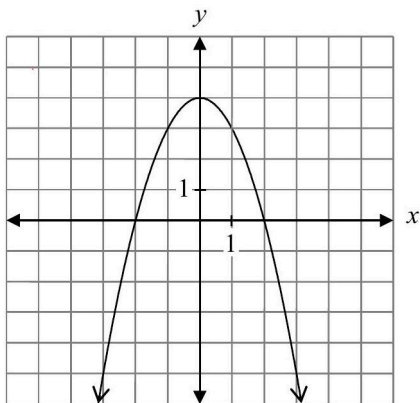
Solution

Correct answer: D

$$f(x) = 2^x - 3$$

Question 24

Given the graph of $y = f(x)$, what is the domain of $\sqrt{f(x)}$?



A.

$$x \in \mathbb{R}$$

B.

$$-2 \leq x \leq 2$$

C.

$$x \leq -2 \text{ or } x \geq 2$$

D.

$$0 \leq x \leq 4$$

Solution

Correct answer: B

$$-2 \leq x \leq 2$$

Question 25

Solve:

$$e^{\ln(5-x)} = 7$$

A.

$$-2$$

B.

$$-\ln 2$$

C.

$$\ln 7 - \ln 5$$

D.

$$\frac{7}{5}$$

Solution

Correct answer: A

$$-2$$

Question 26

One of the factors of $P(x) = x^3 - kx^2 - 7x + 10$ is $(x - 2)$.

Find the value of k .

Solution

Method 1

$$x = 2$$

$$0 = (2)^3 - k(2)^2 - 7(2) + 10$$

$$0 = 8 - 4k - 14 + 10$$

$$0 = 4 - 4k$$

$$4k = 4$$

$$k = 1$$

$\frac{1}{2}$ mark for $x = 2$

1 mark for remainder theorem

$\frac{1}{2}$ mark for solving for k

2 marks

Method 2

$$x = 2$$

2	1	-k	-7	10
		2	$-2k + 4$	$-4k - 6$
	1	$-k + 2$	$-2k - 3$	$-4k + 4$

$\frac{1}{2}$ mark for $x = 2$

$\frac{1}{2}$ mark for synthetic division

$$-4k + 4 = 0$$

$$4k = 4$$

$$k = 1$$

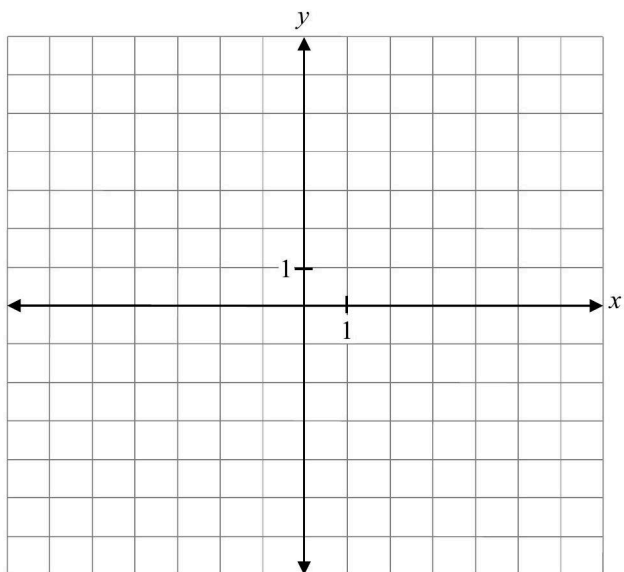
$\frac{1}{2}$ mark for equating the remainder to zero

$\frac{1}{2}$ mark for solving for k

2 marks

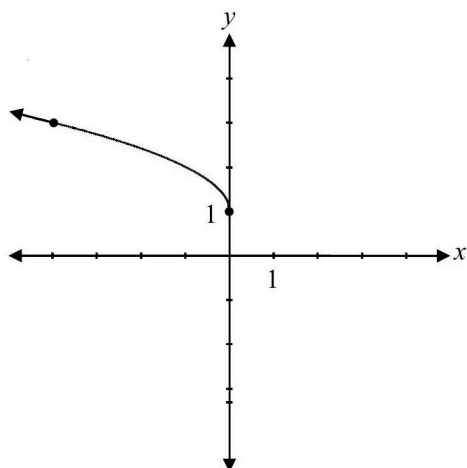
Question 27

a) Sketch the graph of the function $y = \sqrt{-x} + 1$.



b) Determine the value of x when $y = 3$.

Solution



1 mark for general shape

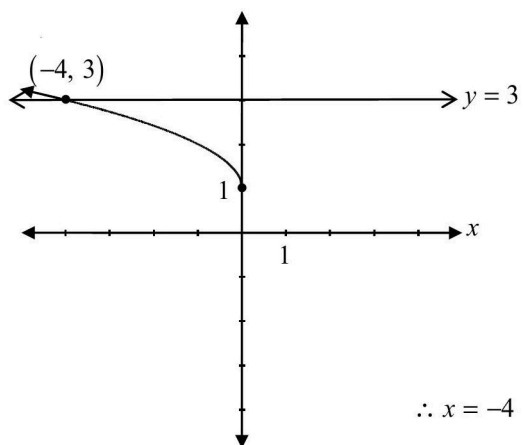
1 mark for horizontal reflection

1 mark for vertical shift

3 marks

b) Determine the value of x when $y = 3$.

Method 1 y



Method 2

$$y = \sqrt{-x} + 1$$

$$3 = \sqrt{-x} + 1$$

$$2 = \sqrt{-x}$$

$$4 = -x$$

$$x = -4$$

1 mark

Question 28

Solve the following equation:

$${}_nP_2 = {}_nC_3$$

Solution

Method 1

$$\begin{aligned}\frac{n!}{(n-2)!} &= \frac{n!}{(n-3)!3!} \\ n!(n-3)!3! &= n!(n-2)! \\ 6 &= \frac{(n-2)!}{(n-3)!} \\ 6 &= \frac{(n-2)(n-3)!}{(n-3)!} \\ 6 &= n-2 \\ 8 &= n\end{aligned}$$

$\frac{1}{2}$ mark for substituting for ${}_nP_2$

$\frac{1}{2}$ mark for substituting for ${}_nC_3$

1 mark for simplification

1 mark for expansion of $(n-2)!$

3 marks

Method 2

$$\begin{aligned}{}_nP_2 &= {}_nC_3; \text{ we know } n \geq 3 \\ \frac{n!}{(n-2)!} &= \frac{n!}{(n-3)!3!} \\ \frac{n(n-1)(n-2)!}{(n-2)!} &= \frac{n(n-1)(n-2)(n-3)!}{(n-3)!6} \quad \text{and } (n-1) \text{ since we know } n \geq 3 \\ &\quad \text{we can divide by } n \\ 6 &= \frac{n(n-1)(n-2)}{n(n-1)} \\ 6 &= n-2 \\ 8 &= n\end{aligned}$$

$\frac{1}{2}$ mark for substituting for ${}_nP_2$

$\frac{1}{2}$ mark for substituting for ${}_nC_3$

$\frac{1}{2}$ mark for expansion of ${}_nP_2$

$\frac{1}{2}$ mark for expansion of ${}_nC_3$

1 mark for simplification

3 marks

Method 3

$$\begin{aligned}{}_nP_2 &= {}_nC_3 \\ \frac{n!}{(n-2)!} &= \frac{n!}{(n-3)!3!} \\ \frac{n(n-1)(n-2)!}{(n-2)!} &= \frac{n(n-1)(n-2)(n-3)!}{(n-3)!3!} \\ 6n(n-1) &= n(n-1)(n-2) \\ 6n(n-1) - n(n-1)(n-2) &= 0 \\ n(n-1)(6 - (n-2)) &= 0\end{aligned}$$

$$n(n-1)(8-n) = 0$$

$$n < 0 \quad n < 1 \quad n = 8$$

$\frac{1}{2}$ mark for substituting for ${}_nP_2$

$\frac{1}{2}$ mark for substituting for ${}_nC_3$

$\frac{1}{2}$ mark for expansion of ${}_nP_2$

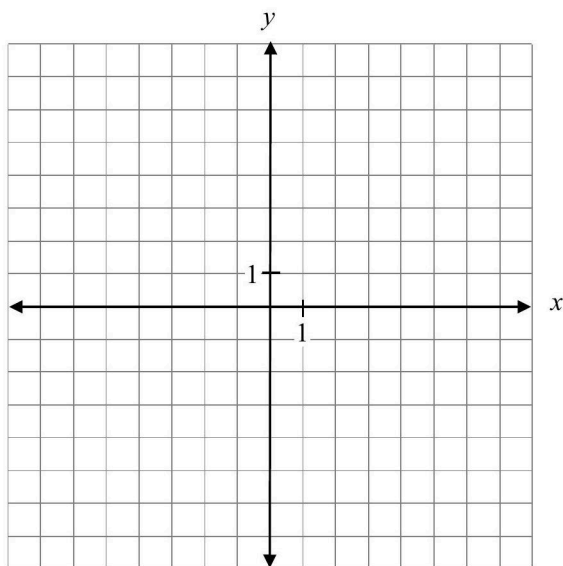
$\frac{1}{2}$ mark for expansion of ${}_nC_3$

1 mark for simplification

3 marks

Question 29

Given $f(x) = x^2 - 1$ and $g(x) = \sqrt{x+1}$, sketch the graph of $y = f(g(x))$ and state its domain.



Domain:

Solution

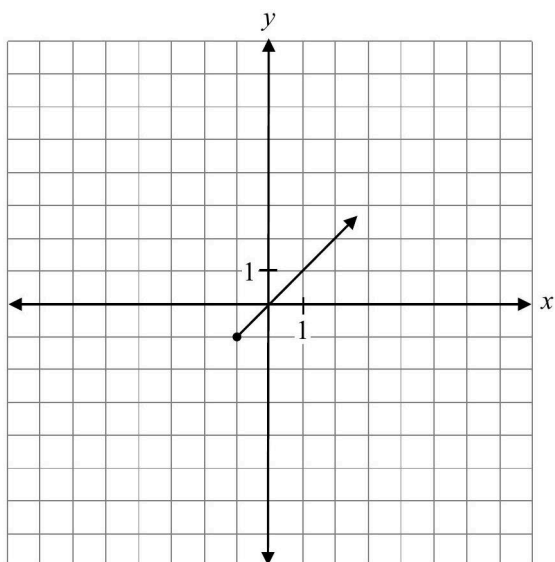
Method 1

$$f(x) = x^2 - 1 \quad g(x) = \sqrt{x+1}$$

Domain: $x \in \mathbb{R}$ Domain: $[-1, \infty)$

$$\begin{aligned} f(g(x)) &= (\sqrt{x+1})^2 - 1 \\ &= x + 1 - 1 \\ &= x \end{aligned}$$

1 mark for determining the function $f(g(x))$



1 mark for graph of composite function

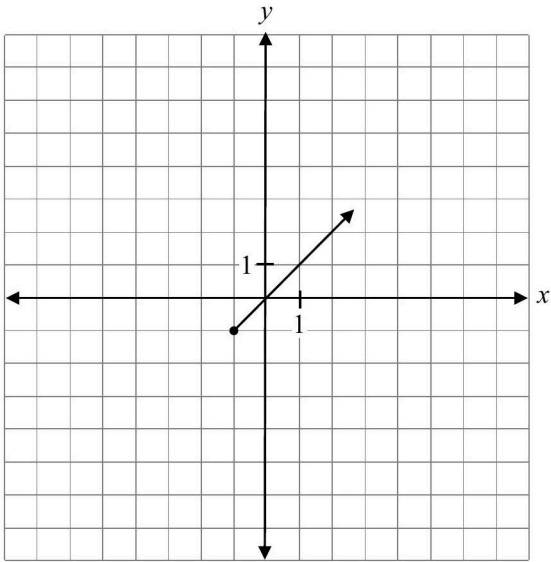
Domain of $f(g(x)) : [-1, \infty)$ or $\{x \mid x \geq -1, x \in \mathbb{R}\}$ 1 mark for restricted domain

3 marks

Method 2

x	$g(x)$	$f(g(x))$
-2		
-1	0	-1
0	1	0
1		1
2		2
3	2	3

1 mark for table of values



1 mark for graph of composite function

Domain of $f(g(x)) : [-1, \infty)$ or $\{x \mid x \geq -1, x \in \mathbb{R}\}$

1 mark for restricted domain

3 marks

Question 30

Write the equation of the horizontal asymptote for the function $f(x) = \frac{x-3}{x-2}$.

Solution

$y = 1$ 1 mark for equation of horizontal asymptote

1 mark

Question 31

The x -intercept of $f(x)$ is 4 and the x -intercept of $g(x)$ is 4 .

Benjamin concludes that the x -intercept of $f(x) + g(x)$ is 8 .

Do you agree with Benjamin? Justify your answer.

Solution

No, I do not agree with Benjamin.

Benjamin added the x -values instead of adding the y -values.

If the x -intercept of $f(x)$ is 4, then $y = 0$.

If the x -intercept of $g(x)$ is 4, then $y = 0$.

$\therefore$ the x -intercept of $f(x) + g(x)$ is 4.

1 mark for justification

1 mark

Question 32

Solve the following equation:

$$2\log_4 x - \log_4(x + 3) = 1$$

Solution

$$2\log_4 x - \log_4(x + 3) = 1$$

$$\log_4\left(\frac{x^2}{x + 3}\right) = 1$$

$$4^1 = \left(\frac{x^2}{x + 3}\right)$$

$$4(x + 3) = x^2$$

$$x^2 - 4x - 12 = 0$$

$$(x - 6)(x + 2) = 0$$

$$x = 6 \quad x = -2$$

1 mark for power rule

1 mark for quotient rule

1 mark for exponential form

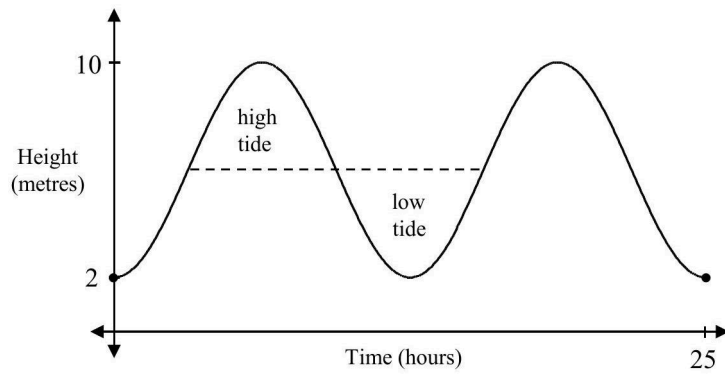
$\frac{1}{2}$ mark for solving for x

$\frac{1}{2}$ mark for rejecting extraneous root

4 marks

Question 33

The following graph represents tidal levels in the Bay of Fundy over a 25 -hour period.



a) What is the average height of the water?

b) What is the period of the graph above?

Explain what the period represents in this situation.

Solution

a) 6 metres

b)

$$\begin{aligned}\text{Period} &= \frac{25}{2} & 1 \text{ mark for period} \\ &= 12.5 \text{ hours}\end{aligned}$$

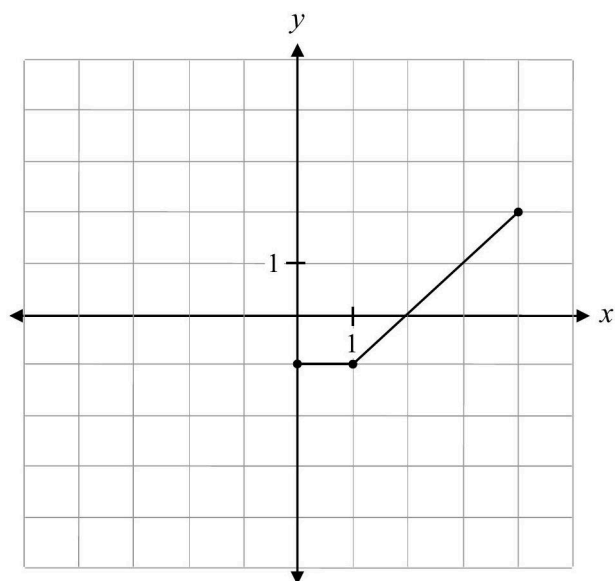
The period represents the time to complete

1 mark for explanation

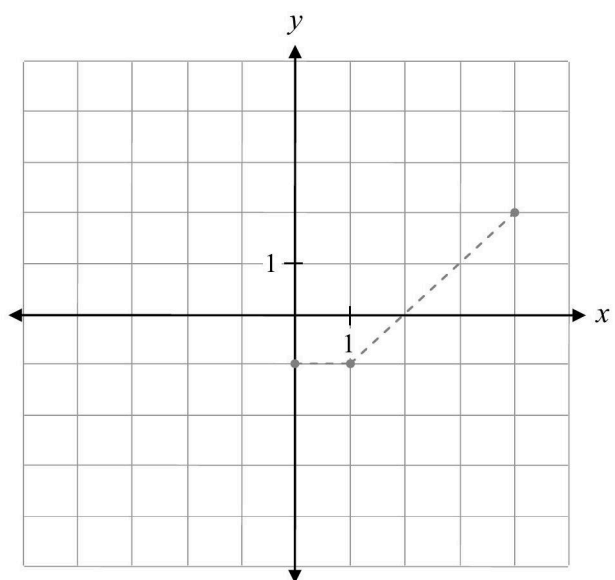
2 marks

Question 34

Given the graph of $y = f(x)$ below,



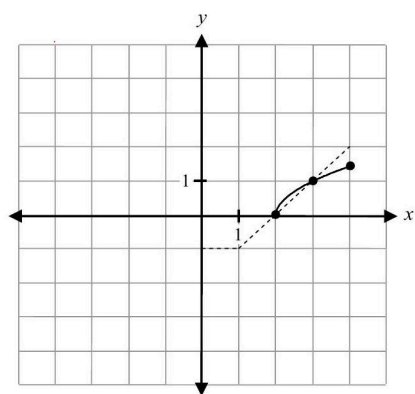
sketch the graph of $y = \sqrt{f(x)}$.



The graph of
 $f(x)$ has
already been
drawn for your
reference.

No marks will
be awarded for
the graph of
 $f(x)$.

Solution



1 mark for restricting the domain

$\frac{1}{2}$ mark for graph above $y = f(x)$ over the range $[0,1]$

$\frac{1}{2}$ mark for graph below $y = f(x)$ over the range $[1,2]$

2 marks

Question 35

When $P(x)$ is divided by $x - 3$, it has a quotient of $2x^2 + x - 6$ and a remainder of 4 .

Determine $P(x)$.

Solution

$$P(x) = (x - 3)(2x^2 + x - 6) + 4$$

or

$$P(x) = 2x^3 - 5x^2 - 9x + 22$$

Question 36

Identify the domain and range of the following function:

$$f(x) = \frac{3}{x^2 + 1}$$

Solution

Domain: $\{x \in \mathbb{R}\}$

1 mark for domain

or

$$(-\infty, \infty)$$

Range: $\{y \in \mathbb{R} \mid 0 < y \leq 3\}$

or

$$(0, 3]$$

1 mark for range

2 marks

Question 37

Evaluate:

$$\csc\left(\frac{11\pi}{6}\right) + \sin^2\left(-\frac{3\pi}{4}\right) + \cos\left(\frac{23\pi}{3}\right)$$

Solution

$$\begin{aligned} &= (-2) + \left(-\frac{\sqrt{2}}{2}\right)^2 + \frac{1}{2} \\ &= -2 + \frac{1}{2} + \frac{1}{2} \\ &= -1 \end{aligned}$$

1 mark for $\csc\left(\frac{11\pi}{6}\right)$ (1/2 mark for quadrant, 1/2 mark for value)

1 mark for $\sin^2\left(-\frac{3\pi}{4}\right)$ (1/2 mark for quadrant, 1/2 mark for value)

1 mark for $\cos\left(\frac{23\pi}{3}\right)$ (1/2 mark for quadrant, 1/2 mark for value)

3 marks

Question 38

Evaluate the coefficient of the term containing x^3 in the expansion of $(1+x)^7$.

Justify your answer.

Solution

Method 1

1

$$111$$

$$1\frac{2}{1}$$

$$1331$$

$$144$$

1 mark for justification

$$1551010$$

$$1615 \quad 20 \quad 15 \quad 6 \quad 1$$

$$1 \quad 7 \quad 21 \quad (35) \quad 35 \quad 21 \quad 7 \quad 1$$

The coefficient of x^3 is 35 .

1 mark for identifying coefficient

2 marks

Method 2

$$t_4 = {}_7C_3(1)^4(x)^3$$

$${}_7C_3 = \frac{7!}{3!4!}$$

$$= \frac{7 \cdot 6 \cdot 5 \cdot 4!}{3!4!}$$

$$= 35$$

1 mark for justification

1 mark for evaluating coefficient

2 marks

Question 39

Which of the following equations could be solved without the use of logarithms?

Without actually solving the problem, explain your choice.

$$4^x = 10^{3x+1} \quad \text{or} \quad \left(\frac{1}{3}\right)^{2x+1} = 27^{4x-1}$$

Solution

$\left(\frac{1}{3}\right)^{2x+1} = 27^{4x-7}$ can be solved without the use of logarithms because $\frac{1}{3}$ and 27 can both be changed to a base of 3 .

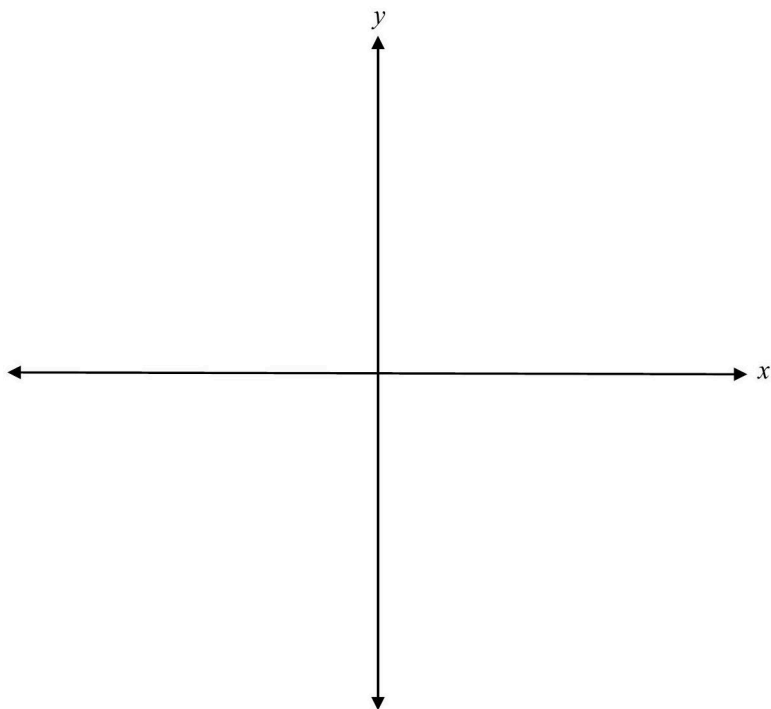
1 mark for explanation

1 mark

Question 40

Sketch the graph of $y = x^3 + x^2 - 5x + 3$ given that one of the x -intercepts is 1 .

Identify the x -intercepts and y -intercept.



Solution

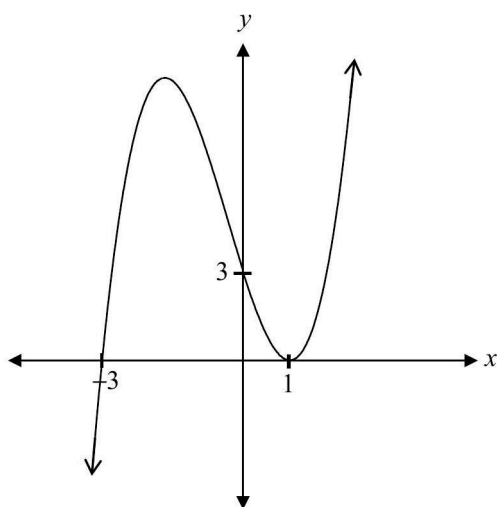
$$x = 1$$

1	1	1	-5	3
		1	2	-3
1	2	-3	0	

$$y = (x - 1)(x^2 + 2x - 3)$$

$$y = (x - 1)(x + 3)(x - 1)$$

$$y = (x + 3)(x - 1)^2$$



1 mark for synthetic division

1 mark for identifying the factors

2 marks for graph ($\frac{1}{2}$ mark for x -intercepts,

$\frac{1}{2}$ mark for multiplicity, $\frac{1}{2}$ mark for

y -intercept, $\frac{1}{2}$ mark for end behaviour)

4 marks

Question 41

If $f(x) = \frac{1}{x-2}$ and $g(x) = x - 2$, what is the domain of $f(x) \cdot g(x)$?

Solution

$$f(x) = \frac{1}{x-2} \quad g(x) = x - 2$$

Domain: $x \in \mathbb{R}, x \neq 2$ Domain: $x \in \mathbb{R}$

Domain of $f(x) \cdot g(x) : \{x \in \mathbb{R} \mid x \neq 2\}$

1 mark for domain of $f(x) \cdot g(x)$

1 mark

Question 42

Given $f(x) = (x + 1)^2$ for $x \leq -1$, write the equation of $y = f^{-1}(x)$.

Solution

Method 1

$$y = (x + 1)^2$$

$$x = (y + 1)^2$$

$$y = \pm\sqrt{x} - 1$$

1 mark for inverse

$\frac{1}{2}$ mark for solving for y

Since the domain of $f(x)$ is $x \leq -1$,

the range of the inverse is $y \leq -1$.

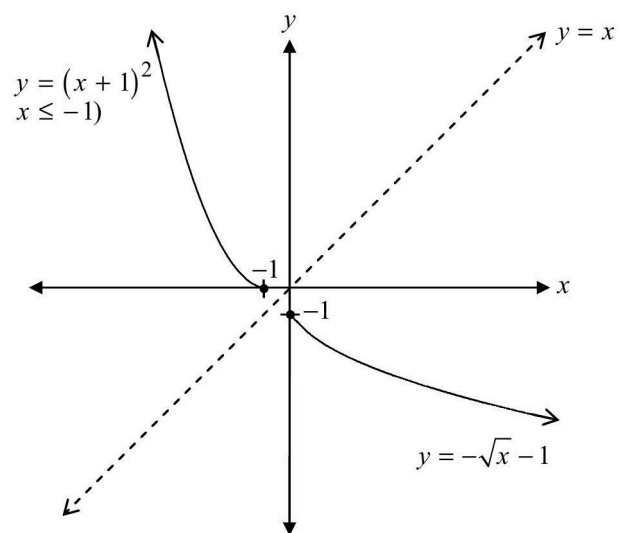
$$\therefore y = -\sqrt{x} - 1$$

$$f^{-1}(x) = -\sqrt{x} - 1$$

$\frac{1}{2}$ mark for rejecting $y = \sqrt{x}$

2 marks

Method 2



1 mark for reflection over the line $y = x$

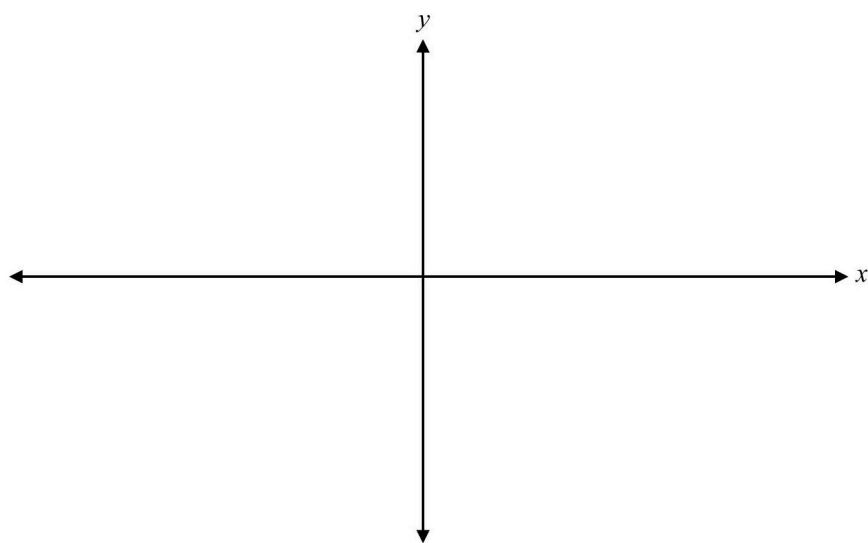
1 mark for correct equation

2 marks

Question 43

Sketch a graph of at least one period of the function $y = 5 \sin[\pi(x + 1)]$.

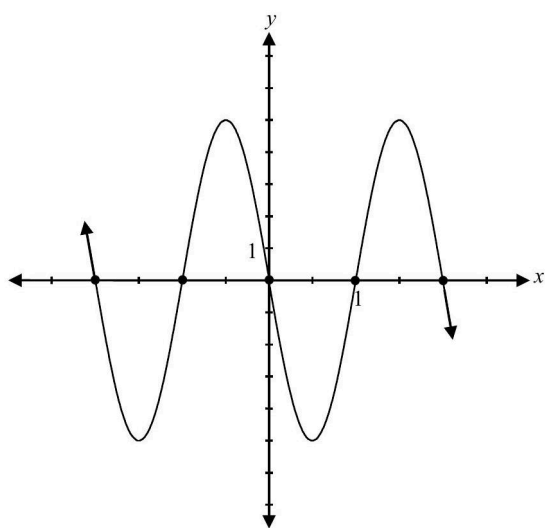
Clearly indicate the x -intercepts.



Solution

$$b = \pi$$

$$\therefore \text{period} = \frac{2\pi}{\pi} = 2$$



1 mark for amplitude

1 mark for horizontal shift

1 mark for period

1 mark for clearly indicating at least two x -intercepts

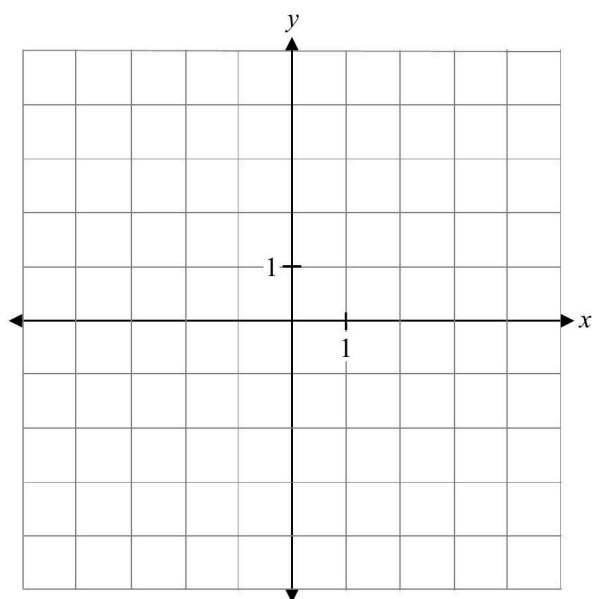
consistent with graph

4 marks

Question 44

Sketch the graph of the following function:

$$f(x) = \frac{x-2}{(2x-3)(x-2)}$$



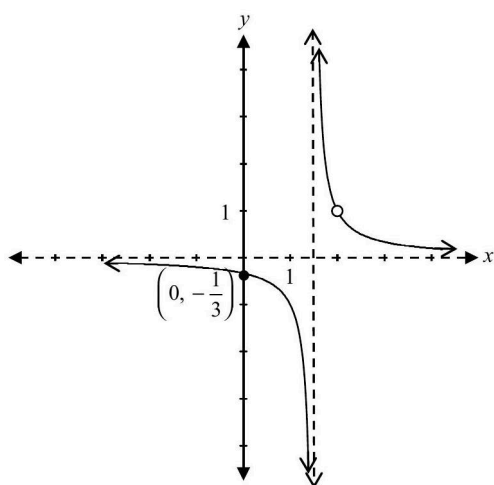
Solution

$$\begin{aligned} f(x) &= \frac{x-2}{(2x-3)(x-2)} \\ &= \frac{1}{2x-3} \quad \text{with a point of discontinuity at } x=2 \end{aligned}$$

point of discontinuity: $f(2) = 1$

$\therefore$ there is a point of discontinuity at $(2, 1)$

$$\begin{aligned} y\text{-intercept: } f(0) &= \frac{0-2}{(2(0)-3)(0-2)} \\ &= -\frac{2}{6} \\ &= -\frac{1}{3} \end{aligned}$$



1 mark for horizontal asymptote at $y = 0$

1 mark for vertical asymptote at $x = \frac{3}{2}$

$\frac{1}{2}$ mark for graph left of vertical asymptote

$\frac{1}{2}$ mark for graph right of vertical asymptote

1 mark for point of discontinuity at $(2, 1)$; ($\frac{1}{2}$ mark

for $x = 2$, $\frac{1}{2}$ mark for $y = 1$)

4 marks